



Gen AI and Agentic AI in Production

Week 4
Day 4

4 Week Journey to Prod-Grade

Week 4: Agentic AI in Production

Multi-Agent
Architecture

Capstone:
Agents on
AWS

Capstone:
Frontend

Observability
Monitoring
Security

Agentic AI
Platforms

Week 3: Azure, GCP, Data Engineering, Vectors, MCP

Cyber AI on
Azure with
MCP

Cyber AI on
GCP with
MCP

Amazon
SageMaker

Data Ingest
S3 Vectors

Data Pipes
with MCP

Week 2: AI Platform Engineering on AWS

AWS
Architecture

The Digital
Twin Mk 2

Amazon
Bedrock

Terraform

GitHub
Actions

Week 1: SaaS Generative AI in Production

First Gen AI
Deployment

Frontend &
Backend

Auth &
Subscription

Healthcare
SaaS App

Deploy to
AWS

Alex - the financial planner



What it is:

SaaS commercial application, financial planner

Done:

Researcher Agent with Data Ingest

Day 1: Database

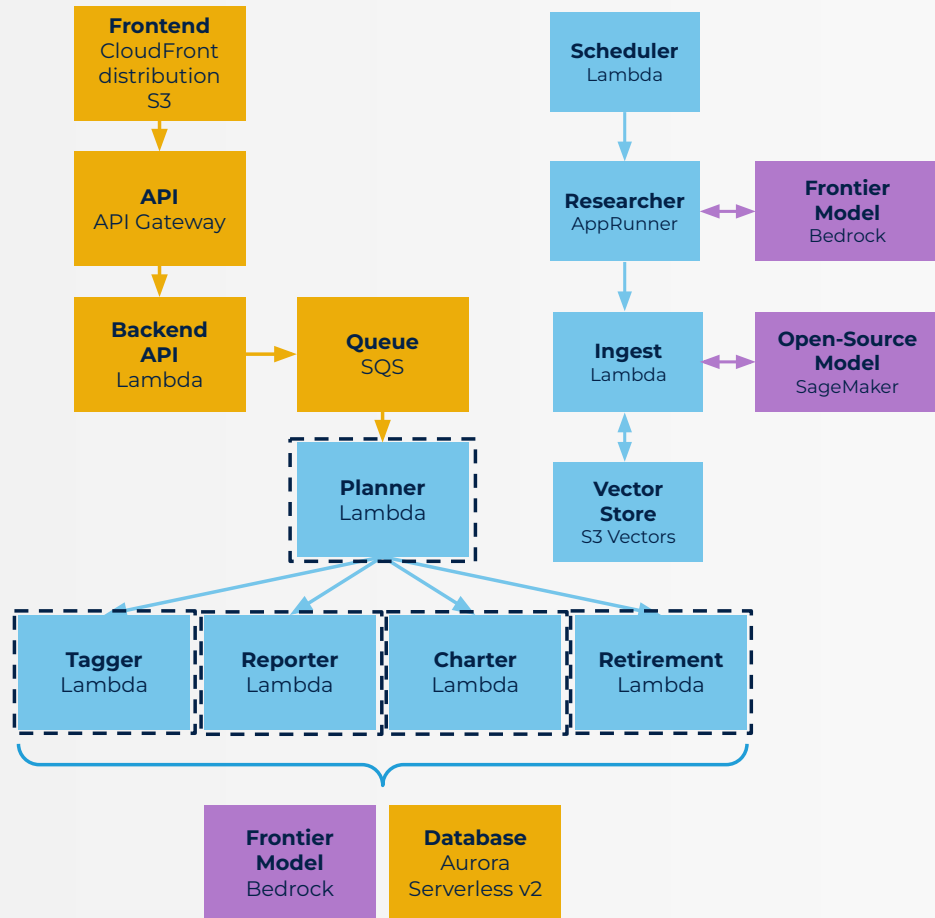
Day 2: Five Agents

Day 3: Frontend + API

Todo:

Day 4: Enterprise Grade

Multi-Agent Architecture



Enterprise Grade



Scalability

Horizontally scaling to meet demand with no downtime



Security

Best practice information security



Monitoring

Best of breed tooling for tracking activities



Guardrails

Check LLM inputs and outputs to protect against LLM weaknesses



Explainability

Understand the rationale for LLM outputs



Observability

Deep insight into LLM behavior as traces and spans

Simon Willison's Lethal Trifecta



Assignment: one of:



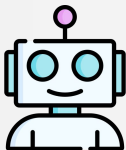
Data Engineering

Actually make the data pipes be relevant to the portfolios



MLOps

Implement the more substantive monitoring and guardrails, track model drift



Agentic AI

Improve the financial planner! Better context engineering and agent interactions

And then submit a PR with a notebook in **production/community_contributions** with a link to your repo

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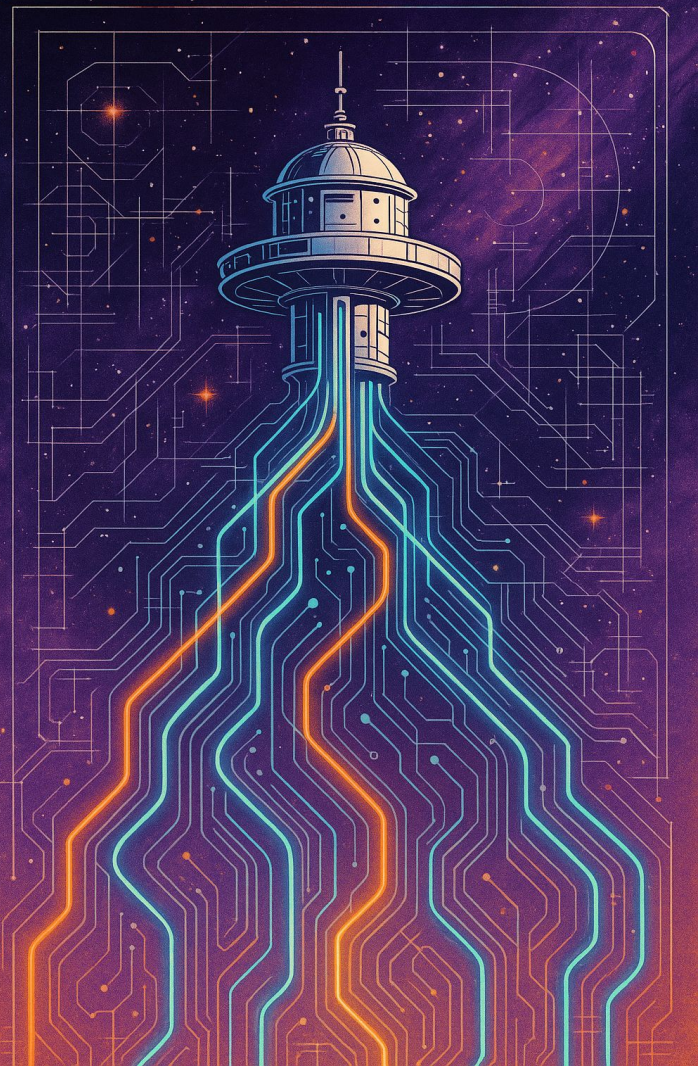
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95%
Complete